Atividade 05 – Dado Eletrônico, Servo Motor e Sensor de Distância

DISCIPLINA	TURMA	PROFESSOR	DATA
Sistemas Embarcados	1ºC2	ROBERTO / RONILDO	28/07/2026

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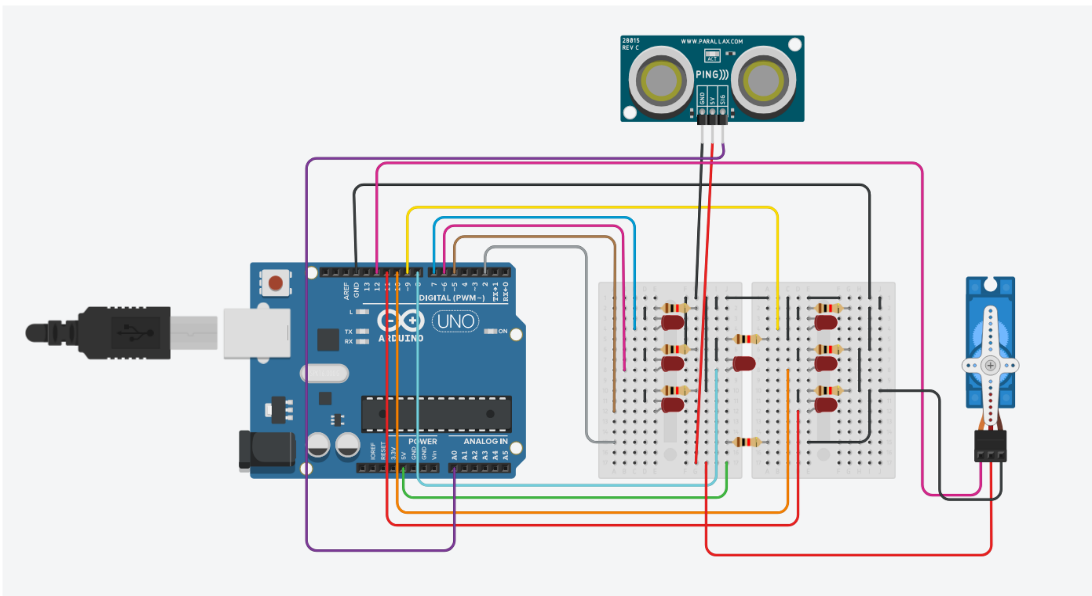
Turma: 1ºC2 — Sistemas Embarcados

1. Montagem do Projeto



2. Código Desenvolvido

```
#include <Servo.h>

long readUltrassonicDistance(int triggerPin, int echoPin)
{
  pinMode(triggerPin, OUTPUT);
  digitalWrite(triggerPin, LOW);
  delayMicroseconds(2);
  digitalWrite(triggerPin, HIGH);
  delayMicroseconds(10);
  digitalWrite(triggerPin, LOW);
  pinMode(echoPin, INPUT);
  return pulseIn(echoPin, HIGH);
}

bool Rolou = false;
float centimetro;
int valor;
Servo servo1;

void apaga(){
  digitalWrite(11, LOW);
```

```
  digitalWrite(10, LOW);
  digitalWrite(9, LOW);
  digitalWrite(8, LOW);
  digitalWrite(7, LOW);
  digitalWrite(6, LOW);
  digitalWrite(5, LOW);
}

void setup()
{
  randomSeed(analogRead(A1));
  pinMode(11, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(9, OUTPUT);
  pinMode(8, OUTPUT);
  pinMode(7, OUTPUT);
  pinMode(6, OUTPUT);
  pinMode(5, OUTPUT);
  pinMode(4, OUTPUT);
  pinMode(3, OUTPUT);
  apaga();
  servo1.attach(12);
  servo1.write(0);
}

void sorteio(){
  valor = random(1,7);
  if (valor == 1){
    digitalWrite(8, HIGH); }
  if (valor == 2){
    digitalWrite(7, HIGH); digitalWrite(11, HIGH); }
  if (valor == 3){
    digitalWrite(7, HIGH); digitalWrite(11, HIGH);
    digitalWrite(8, HIGH); }
  if (valor == 4){
    digitalWrite(5, HIGH); digitalWrite(11, HIGH);
    digitalWrite(7, HIGH); digitalWrite(9, HIGH); }
  if (valor == 5){
    digitalWrite(5, HIGH); digitalWrite(11, HIGH);
    digitalWrite(7, HIGH); digitalWrite(9, HIGH);
    digitalWrite(8, HIGH); }
  if (valor == 6){
    digitalWrite(10, HIGH); digitalWrite(11, HIGH);
    digitalWrite(7, HIGH); digitalWrite(9, HIGH);
    digitalWrite(6, HIGH); digitalWrite(5, HIGH); }

  delay(4000);
  apaga();
}

void s_motor(){
  servo1.write(90);
  delay(2000);
  servo1.write(0);
  delay(1000);
}

void loop() {
  centimetro = 0.01723 * readUltrassonicDistance(A0, A0);
  if (centimetro < 40){
    if (!Rolou){
      Rolou = true;
      sorteio();
      if (valor % 2 == 0) {
        s_motor();
      }
    }
```

```
  } else {
    Rolou = false;
  }
}
```